Session 4: Solutions and facilitation notes

Foundations of Clinical Research · Week 4 · Saturday 19:30-22:30, online · Neudata · Instructor copy

Answer keys for `Practicals/s4_exercises.md` (the Session 4 exercise sheet, sent to participants as a PDF before the session). All numbers were checked with `Resources/tools/2x2_calculator.xlsx`. Share this key with participants after the session, for the homework and self-study items.

In-session exercises (Parts A, C and E) are whole-class tasks: show the item on the shared screen, run a poll, take one volunteer's answer from the chat, and type the class answer on the shared screen.

Part A: Spot the bias (whole class, 5 minutes)

#	Main bias	Why	One way to reduce it (any of these)
1	**Recall bias** (a type of information bias)	Families of patients who died search their memory harder than families of survivors, so exposures look more common in cases.	Use exposure data recorded before the outcome was known (charts, prescription records); ask both groups the same structured questions; blind interviewers to case or control status.
2	**Observer bias** (information bias)	A soft, judgement-based outcome assessed by someone who knows the group.	Blinded outcome assessor; objective, pre-defined discharge criteria; a harder outcome (e.g. actual days to ICU discharge).
3	**Selection bias**	Requiring 3 days in the ICU leaves out the sickest patients who died early (and the mildest who left early); the sample no longer represents all patients with sepsis. This is exactly the limitation of Paper 2 (Hyun).	Include all eligible patients from admission; report how many were excluded and why; state the limitation and who the result applies to.
4	**Confounding (by indication)**	Lactate (severity) leads to more fluid AND independently predicts death. The crude association overstates (or creates) the effect of fluid.	Randomise; or measure severity at baseline (lactate, SOFA score, vasopressor dose) and restrict, match or adjust; say "associated with", not "increases".

Part B: Confounding or effect modification? (homework)

#	Answer	Why	What to report
1	**Confounding** (by indication: lactate)	Both strata give RR 1.00. They agree with each other but not with the crude RR 1.80. Lactate drove the choice of drug and predicts death.	The lactate-adjusted (or stratified) estimate, RR about 1.0. The crude 1.80 is misleading.
2	**Effect modification** (by age)	The strata differ from each other: RR 0.50 under 65, 0.95 at 65 and over.	Each age group separately (ideally a pre-specified subgroup with a test for interaction). Do not report only the average 0.70.
3	**Neither**	The strata agree with each other (0.80, 0.79) and with the crude result (0.80).	The overall RR 0.80; the shift neither confounds nor modifies the effect.

Typical error: calling scenario 2 "confounding" because the numbers change. Ask: *"Do the two strata agree with EACH OTHER?"* If not, it is a finding about who benefits, not a bias to remove.

Part C: Two 2x2 calculations (whole class, 15 minutes)

About 7 minutes per calculation: poll each answer, then type the table into the calculator on the shared screen.

Calculation 1: fluid trial (200 per group)

	Died: YES	Died: NO	Total
Restrictive	a = 20	b = 180	200
Usual care	c = 30	d = 170	200

Measure	Working	Answer
Risk, restrictive	20 / 200	**0.10 (10%)**
Risk, usual care	30 / 200	**0.15 (15%)**
RR	0.10 / 0.15	**0.67**
RD	0.10 − 0.15	**−0.05 (5 percentage points fewer)**
NNT	1 / 0.05	**20**
OR	(20 × 170) / (180 × 30) = 3400 / 5400	**0.63**

- **Why is the NNT different from the worked example?** Same RR (0.67), but the baseline risk is lower (15% instead of 30%), so the absolute difference halves (5 instead of 10 points) and the NNT doubles (20 instead of 10).
- **Model sentence:** "With restrictive fluids, 10 in 100 patients died by day 28 compared with 15 in 100 with usual care: about one death prevented for every 20 patients treated."

Common errors: dividing by the wrong total (50 or 400); writing RD as 5 instead of 0.05 and then NNT = 0.2; table drawn with the outcome in rows.

Calculation 2: fluid case-control study

	Cases (new dialysis)	Controls (no dialysis)
Positive fluid balance on day 3	a = 24	b = 16
Zero or negative balance	c = 16	d = 64

- **OR** = (24 × 64) / (16 × 16) = 1536 / 256 = **6.0**
- **Why no risk, RR or NNT?** The investigators chose how many people with and without dialysis to include (40 and 80, a 1:2 ratio). The proportion needing dialysis in the study is fixed by design, not by nature, so a "risk" calculated from the rows is meaningless. The OR is valid because it does not depend on the sampling ratio.
- **Reverse causation?** Yes, quite possibly: failing kidneys make less urine, and less urine out makes the fluid balance positive. The kidney injury may have come first.
- **Model sentence:** "Patients who needed dialysis had about six times the odds of a positive fluid balance on day 3, compared with similar ICU patients who did not. This observational study cannot show that the fluid caused the kidney failure: sicker patients receive more fluid, and failing kidneys themselves make the balance positive."

Calculator caution: typed into the Association sheet, these numbers give "risks" of 60% and 20%. Use this to reinforce the red warning on the sheet: for case-control data, read only the OR.

Extra practice (self-study): pressure-relieving mattress trial

	Pressure injury: YES	NO	Total
New mattress	a = 10	b = 190	200
Standard mattress	c = 20	d = 180	200

Risks **5%** and **10%**; RR **0.50**; RD **−5 percentage points**; NNT **20**; OR = (10 × 180) / (190 × 20) = **0.47**. The OR is fairly close to the RR because the outcome is uncommon (5-10%). Model sentence: "With the new mattress, pressure injuries were halved (5% vs 10%): about 1 injury prevented for every 20 patients who use it."

Part D: PPV in two settings (homework)

Setting 1: ICU, prevalence 20% (500 people)

	Disease present	Disease absent	Total
Test positive	TP = 90	FP = 80	170
Test negative	FN = 10	TN = 320	330
Total	100	400	500

PPV = 90 / 170 = **52.9%** · NPV = 320 / 330 = **97.0%**

Setting 2: screening, prevalence 2% (500 people)

	Disease present	Disease absent	Total
Test positive	TP = 9	FP = 98	107
Test negative	FN = 1	TN = 392	393

	Disease present	Disease absent	Total
Total	10	490	500

PPV = 9 / 107 = **8.4%** · NPV = 392 / 393 = **99.7%**

1. **Why different?** With a rare disease, even a modest false-positive rate (20%) applied to the many healthy people produces far more false positives (98) than true positives (9). Sensitivity and specificity did not change; the population did.
2. **LR+** = 0.90 / 0.20 = **4.5** in both settings. Likelihood ratios do not depend on prevalence.
3. **Prevalence 50%:** TP 225, FN 25, FP 50, TN 200, giving PPV 225 / 275 = **81.8%**, NPV 200 / 225 = **88.9%**.

These match the slide "Same test, different setting" (4.3) and the prevalence table on the Diagnostic sheet of the 2x2 calculator.

Part E: Read a forest plot and choose the risk-of-bias tool (whole class, 8 minutes)

Numbers are computed by inverse-variance (fixed-effect) pooling of the log risk ratios in `figs_sat2.py`.

Trial	RR (95% CI)	Weight
Trial 1	0.80 (0.41-1.56)	10%
Trial 2	0.83 (0.54-1.28)	25%
Trial 3	1.14 (0.46-2.85)	5%
Trial 4	0.78 (0.55-1.10)	38%
Trial 5	0.74 (0.45-1.23)	18%
Trial 6	0.62 (0.23-1.69)	5%
Pooled	**0.79 (0.64-0.98)**	I^2 = 0%

1. **Most weight:** Trial 4 (38%). It is the largest trial with the most deaths (103), so it has the narrowest CI.
2. **Any single trial clear?** No. Every individual CI crosses 1.
3. **Pooled:** the diamond does not cross 1. *"Across these six trials, a restrictive strategy was associated with about 21% lower relative risk of death (RR 0.79, 95% CI 0.64 to 0.98); the upper limit is close to no effect."* (For randomised trials "reduced" is acceptable; the point is to give the CI.) Type the best sentence from the chat on the shared screen.
4. **Trial 3 (self-study):** no great concern. It is small (5% weight, 15 deaths) with a very wide CI that overlaps all the others; I^2 = 0% means the spread is compatible with chance.

Choosing the tool

#	Study	Tool
a	Paper 1, Boulet 2024 (RCT)	**RoB 2**
b	Paper 2, Hyun 2023 (cohort, exposure)	**Newcastle-Ottawa** (or **ROBINS-E**)

#	Study	Tool
c	Systematic review and meta-analysis	**AMSTAR 2**
d	Before-after fluid protocol study	**ROBINS-I** (non-randomised study of an intervention)
e	Point-of-care vs laboratory lactate	**QUADAS-2** (diagnostic accuracy)
f	Cluster-randomised fluid-charting trial	**RoB 2** (its cluster-randomised version)

Common error: choosing ROBINS-I for Hyun. Accept it with discussion (fluid balance is partly a treatment decision), but the usual choice for an exposure cohort is Newcastle-Ottawa or ROBINS-E.

Part F: Proposal builder 4 (what to model live)

The instructor fills this in on the shared screen during the first 15 minutes, thinking aloud. Each participant then writes the same rows for their own question. (Rows 1-3, covering design, setting and population, are in `s3_solutions.md`, Part C.)

Section	Model answer (observational version of the running question)
Exposure and comparator	Exposure: positive cumulative fluid balance (ml/kg) at 72 h after ICU admission. Comparator: zero or negative cumulative balance at 72 h.
Outcomes	Primary: death from any cause by day 28 after ICU admission (hospital record; phone call if discharged). Secondary: acute kidney injury by day 7 (KDIGO criteria); days alive and free of mechanical ventilation by day 28; ICU length of stay.
Biases and mitigation	Confounding by indication (sicker patients receive more fluid): record SOFA score, lactate, vasopressor dose, age and comorbidity at admission and adjust or match. Selection bias (requiring 72 h of ICU stay excludes early deaths): state it as a limitation and report how many were excluded and why. Information bias (incomplete fluid charts, missing night-time urine output): pre-specified definitions, two data collectors with a double-entry check on a 10% sample.

Claim: "associated with" (observational design).

Common problems to look for in the chat and in the drafts

1. No comparator ("we will study patients who receive X").
2. Three "primary" outcomes, or a primary outcome without a time point.
3. Observational design with no mention of confounding.
4. A bias named with no concrete mitigation ("we will be careful").